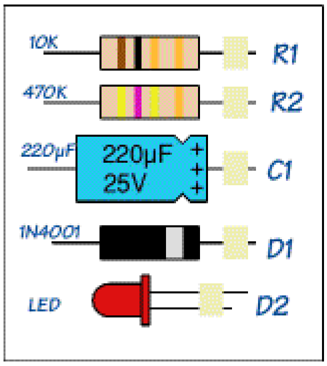
Starter kit of components

Also see: Tools for electronics | Making a workbench



If you are new to electronics and would like to try adapting published projects, or designing and building your own circuits, you need to have a small stock of components available. However, there is a very wide range of components and it can be difficult to know which ones you really need! I hope the list below will help you choose a sensible selection which is within your budget. Remember that circuits built on breadboard can be dismantled after use and the components re-used.

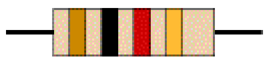
It is usually cheapest to buy components by mail order and several suppliers are listed on the Links page. Send for a catalogue first, even if you have to pay for it, because most include a great deal of useful information as well as listing part numbers and prices. Kits of assorted components may be available and this is a great way to start if you can afford the initial cost. Remember that you will need to organise storage of the components!

Rapid Electronics stock a wide range of components and they have kindly allowed me to use their photographs on this page.

Essential components

These are the components used in most projects. The individual components are quite cheap, but the total cost of the set will be significant! One way to spread the cost is to add a few items from this list every time you buy the components for a particular project. Click on the titles for further information.

Resistors



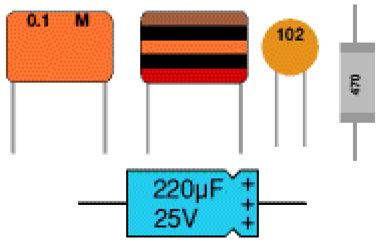
0.25W carbon film resistors are the cheapest type. Choose ones with 4-band colour codes because these are easier to read (the precision of 5-band codes is unnecessary).
Ideally you need a good selection of values over the range 100Ω to 1MΩ such as the E6 or E12 series, but that is a large number of resistors! As a minimum I suggest: 470*, 1k*, 2k2, 4k7, 10k*, 22k, 33k, 47k, 100k, 220k, 470k and 1MΩ. Buy at least 10 of each value and 20 of those marked *. The 470Ω resistors are for use with LEDs, even if a project specifies a slightly different value.

Resistors may be combined in series and parallel to obtain extra values, for example 100kΩ and 220kΩ in series is 320kΩ which is close enough to 330kΩ.

Capacitors

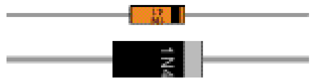
Low values: 0.01µF and 0.1µF metallised polyester, 10 of each.

High values: 1µF 63V, 10µF 25V, and 100µF 25V electrolytic with radial leads, 10 of each; 220µF 25V and 470µF 25V electrolytic with axial leads, 3 of each.



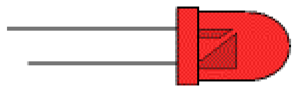
Diodes

1N4148 signal diode and 1N4001 rectifier diode, 5 of each.

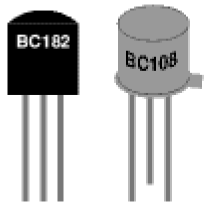


LEDs

Red, yellow and green 5mm standard LEDs, 10 of each.



Transistors

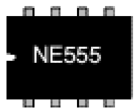


About 5 **general purpose, low power, NPN transistors**. These should have a maximum collector current (Ic max) of 100mA, and a minimum current gain (h_{FE} min) of 200.
For example: BC548B (BC108 equivalent).

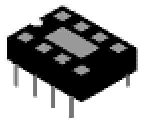
About 5 **general purpose, medium power, NPN transistors**. These should have a maximum collector current (Ic max) of 1A, and a minimum current gain (h_{FE} min) of 30.
For example: BC639 (BFY51 equivalent).

Integrated circuits ('chips') and holders

NE555 timer IC, at least 3 (10 if you plan to solder projects). It is not worth ordering other ICs at this stage unless you know they are needed for some of the projects you wish to try.



If you are planning to solder circuits on stripboard or PCB you will also need 8-pin, 14-pin and 16-pin DIL sockets (chip holders), at least 10 of each.



Variable resistors

Presets are cheaper than standard variable resistors but most have pins which are too large for breadboards. For breadboard circuits is is probably best to buy standard variable resistors and solder short single core 1/0.6mm wires onto them.

The useful values are: 10k LIN, 100k LIN and 1MΩ LIN, buy 2 of each. A 1MΩ LOG potentiometer is useful too. Knobs are optional because it is easy to turn the spindles by hand. If you buy presets the horizontal style are best, all presets are LIN.

Standard Variable Resistor
Photograph © Rapid Electronics

Battery clips

Photograph © Rapid Electronics

Clip for a 9V PP3 battery, buy 3 (or 10 if you plan to solder projects). Remember to buy a battery too!

Wire

Red and black 7/0.2mm stranded wire, one colour of single-core 1/0.6mm wire, 10m (or a reel) of each. If you are planning to build projects on breadboard buy extra colours of the single-core wire, including red and black.

Crocodile clips

Buy at least one standard crocodile clip to use as a heat sink when soldering. Miniature red and black crocodile clips (buy about 10 of each) are useful for making your own test leads using 7/0.2mm stranded wire.

Photographs © Rapid Electronics

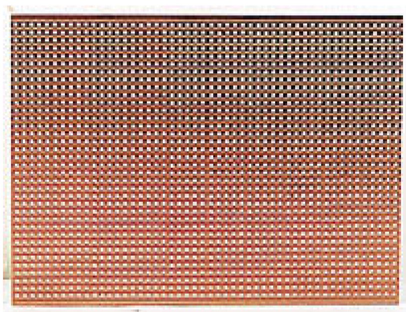
Switches

Photographs © Rapid Electronics

Switches are not essential for breadboard circuits because you can make or break links with pieces of wire. The on/off switch from soldered projects can also be omitted if you are willing to unclip the battery instead.

If you wish to buy a few switches the most useful types are push-to-make and miniature SPDT toggle switches, 3 of each.

Stripboard

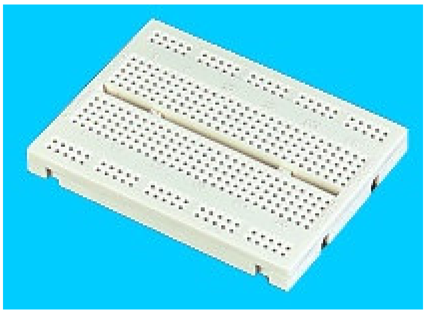


Photograph © Rapid Electronics

Buy a large sheet (or two) and cut it up as required. You can cut it neatly to size using a junior hacksaw, cutting along the lines of holes is easiest. For quickness you can break it over the edge of a workbench along the lines of holes - take care though because this needs a fairly large force and the edges will be rough. You may need to use a large pair of pliers to nibble away any jagged parts.

Avoid handling stripboard that you are not planning to use immediately because sweat from your hands will corrode the copper tracks and this will make soldering difficult unless you clean the board first.

Breadboard



Photograph © Rapid Electronics

A small breadboard (such as the Protobloc 1 shown in the picture) is suitable for simple circuits with up to two ICs, but if you intend to build more complex circuits such as counters it is best to buy a larger breadboard (such as the Protobloc 2).

Breadboards do not require soldering so the components used on them can be re-used many times. They are ideal for testing your own circuit designs and trying out ideas such as adapting a published project.

Other components to consider

Buy these if you can afford them, or wait until they are needed for a project.

- A light dependent resistor (LDR), ORP12 or NORPS12.
- A thermistor, miniature disc type, NTC about 5kΩ@25°C.
- A piezo transducer with flying leads.
- A buzzer or bleeper.
- A miniature loudspeaker 8Ω.

Miniature piezo transducer
Photograph © Rapid Electronics

Storage systems for components

You can store all your components in a single container, such as a plastic food box, but as you accumulate more items it will become increasingly difficult to find the smaller components! A cheap solution is to organise the parts into small snap-top plastic bags which can be labelled, in fact you may find that some components are supplied like this.

Snap-top plastic bags

Drawer cabinet

Photographs © Rapid Electronics

Probably the best storage system is a cabinet of plastic drawers. These can be expensive, but you do not need many drawers because there is no need to have a drawer for every single component value. Many parts can be grouped together, such as decades of resistor values. For example you could organise a 15-drawer cabinet like this:

1. Resistors 10Ω+ (third band **black**) only a few, but they tend to be large
2. Resistors 100Ω+ (third band **brown**)
3. Resistors 1kΩ+ (third band **red**)
4. Resistors 10kΩ+ (third band **orange**)
5. Resistors 100kΩ+ (third band **yellow**)
6. Resistors 1MΩ+ (third band **green**) and 10MΩ (third band **blue**)
7. Presets, also variable resistors if they will fit in the drawer
8. Capacitors low values, less than 1µF

9. Capacitors electrolytic 1µF+
10. Diodes and transistors
11. LEDs and lamps (also LED clips and lampholders)
12. ICs (chips) and their holders (DIL sockets)
13. Switches and relays
14. Connectors (crocodile clips, plugs and sockets)
15. Other components (battery clips, piezo transducers, LDRs, thermistors)

Rapid Electronics stock a wide range of components and they have kindly allowed me to use their photographs on this page. We buy most of our components and tools for the Electronics Club from Rapid Electronics. The photographs are from their Image Gallery CD-ROM.

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